POAMS: LECTURE 19
THE INCORPORATION OF SPIN IN ORBITAL MOTION

In this lecture, we continue to discuss, in Normal Realist terms, the effect of spin on the orbital parameters of free-moving particles and hence, on the value of G. Recall that in the POAMS approach, it is not just orbital angular momentum which is holistically conserved but, rather, the total angular momentum of a particle, i.e. orbital plus spin. This commits us to predictions that are distinct from those of Newtonian theory. This lecture is based on some of Chapter 7 of 'Light-speed....'.

1. Introduction [Recap]

Consider, once again, the case of a freely moving particle, P, of mass m, in a closed orbit about a body of mass M, with $m \ll M$. If P spins (rotates) on its axis while orbiting, then according to POAMS, since any orbital angular momentum is balanced against the total momentum of the system, introducing this spin has a direct and immediate effect on the system as a whole. In other words, POAMS postulates that P's spin will cause it take up a natural orbit different from that which it would follow if it were not spinning.

Although that we have stated in the last lecture that the orbital kinetic energy, K, in P's natural orbit, when spin effects are taken into account, depends on both P's spin kinetic energy, K_s, and the spin-less orbital kinetic energy K_o, we have not yet shown how K may be calculated from K_s and K_o. *From now on*, we shall only consider the case in which P is an *axially symmetric* particle with *constant angular speed* in a *circular* orbit about the body B. In this special case, it follows from equations (7) and (9) of Lecture 18 [$L = \frac{2K}{\omega}$, $S = \frac{2K_S}{\omega}$] that K_o and K_s are respectively proportional to the magnitude of the (spin-less) orbital angular momentum, L, and the magnitude of the P's spin angular momentum, S. In the case of no spin, it follows from equation (7) of lecture 18 that

$$K_o = \frac{1}{2}\omega_o L \quad (1)$$

where ω_o is P's constant orbital angular speed. For a spinning particle P with constant spin angular speed ω, it follows from equation (9) of lecture 18 that

$$K_s = \frac{1}{2}\omega S \quad (2)$$

[*End of recap*]

The maximum possible effects due to spin occur when the particle P spins in either the same or the opposite direction to that of its orbital motion. When P spins in the *same direction* as its orbital motion, its spin angular momentum vector, **S**, points in the same direction as its angular momentum vector, **L**. In this case, the magnitude of the total angular momentum, $\|\mathbf{L} + \mathbf{S}\|$, is $L + S$ and the same is true for any multiples of **L** and **S**. In keeping with equations (1) and (2), POAMS postulates that in this case, the total orbital kinetic energy is

$$\mathcal{K} = K_o + K_s \quad (3)$$

Similarly, when P spins in the *opposite direction* to that of its orbital motion, **S** points in the direction opposite to that of **L** and, in this case, the magnitude of the total angular momentum, $\|\mathbf{L} + \mathbf{S}\|$, is $|L - S|$. We then postulate that in this case,

$$\mathcal{K} = |K_o - K_s| \quad (4)$$

For orbital motion on the macro-scale it follows from physical considerations that $K_s \ll K_o$ so that the case of $\mathcal{K} = 0$ implied by equation (4) does not arise.

In general, if P's spin angular momentum vector, **S**, is inclined at an angle θ to its orbital angular momentum vector, **L**, *i.e.*, if P's axis of rotation is inclined at this angle to the normal of the orbital plane, then the magnitude of the total angular momentum is given by

$$\|\mathbf{L} + \mathbf{S}\|^2 = L^2 + S^2 + 2LS\cos\theta$$

[*see endnote* (*) *for details*]. POAMS then postulates, from equations (1) and (2), that in this general case,

$$\mathcal{K}^2 = K_o^2 + K_s^2 + 2K_oK_s\cos\theta \qquad (5)$$

In particular, if *P*'s spin angular momentum vector lies in the plane of its orbit, so that **S** is perpendicular to **L**, then

$$\mathcal{K}^2 = K_o^2 + K_s^2 \qquad (6)$$

With our hypothesis, this is clearly the minimum possible effect due to spin [*see endnote* (**) *for more details*].

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2. <u>The effect of spin in POAMS: 'attraction' and 'repulsion'</u>

In the previous lecture, I indicated how, in an isolated two-body system, the parameters of a freely moving particle, *P*, in its natural closed orbit about the centre-of-mass, *O*, of a body *B*, are affected by its spin angular momentum and hence by its spin kinetic energy. It follows that *P*'s spin affects either its relative 'attraction' or 'repulsion towards *O* and hence *B*.

Once again, we shall only consider the case in which *P* is *axially symmetric* with *constant spin angular speed* and follows a natural *circular* orbit about *O*. Recall that in this case, the speed, v_s, of *P* in its circular orbit, adjusted for its spin (*i.e.*, *P*'s *effective orbital speed*), is given by equation (15) of Lecture 18, that is,

$$v_s^2 = \frac{2}{m}\mathcal{K} \qquad (7)$$

where *m* is the mass of *P* and $\mathcal{K}$ is the combined orbital kinetic energy of *P* taking the spin of *P* into account. Then the radius, r_s, of *P*'s orbit, taking its spin into account, is given by equation (16) of Lecture 18, *i.e.*,

$$r_s = \frac{L}{mv_s} \qquad (8)$$

where L is constant magnitude of P's conserved orbital angular momentum. It follows by equations (7) and (8) that

$$r_s = \frac{L}{\sqrt{2m}} \mathcal{K}^{-1/2} \qquad (9)$$

Hence, with constant L and m, r_s is proportional to $1/\mathcal{K}^{1/2}$. Now if P spins in the *same* direction as that of its orbital motion, then POAMS postulates that its combined orbital kinetic energy, $\mathcal{K}$, is given by equation (3), so that $\mathcal{K} > K_o$. It then follows by equation (9), with L fixed, that an *increase* in P's orbital kinetic energy produces a *decrease* in the radius of P's natural orbit about B. Hence, in the Newtonian language of 'attraction' and 'repulsion', P is 'more attracted' to B when spinning compared to when it is not. The greater the spin kinetic energy of P, the smaller the value of r_s and so the greater the 'attraction' of P towards B. This makes physical sense if we note that in this case, P is spinning with its orbital motion and so can be thought of as 'tumbling towards' B in its orbit.

If instead, P spins in the opposite direction to that of its orbital motion, then the total orbital kinetic energy is given by equation (4). In general, for macroscopic orbits, $K_s \ll K_o$, so that in this case, $\mathcal{K} < K_o$. Then by equation (9), the presence of P's spin kinetic energy produces an increase in the radius of P's orbit, so that, in Newtonian language, P is 'less attracted' to B when spinning compared to when it is not. This makes physical sense if we note that in this case, since P is spinning in the direction opposite to that of its orbital motion, P is resisting its natural spin-less orbit in some way.

In the general case in which P's axis of rotation is inclined at an angle θ to the normal of the orbital plane, POAMS postulates that its combined orbital kinetic energy, $\mathcal{K}$, is given by equation (5). Then for $0 < \theta < 90^0$, $\mathcal{K} > K_o$, and if $90^o < \theta < 180^o$, then $\mathcal{K} < K_o$. It then follows by equation (9) that there is an increase in P's orbit radius in the first case and a decrease in P's orbit radius in the second.

Of course, in general, on the macroscale, the effect of the spin of a particle on its natural orbit will be tiny, although measurable. However, where K_s is very much larger that K_o, as appears to be the case on the *microphenomenal scale*, which will be discussed in the next lecture, it follows that the differences between the combined orbital kinetic energy,

$\mathcal{K}$, and K_o are very pronounced. In addition, in this case it follows by equation (9) that the result of any very large increase in K_s, and hence $\mathcal{K}$, must necessarily decrease the orbit radius. This is *regardless* of the orientations of the orbital and spin angular momentum vectors where the effect of the first is swamped by that of the second.

It needs to be stressed that in the cases discussed above, we are considering an idealised situation in which P is *spinning* in its orbit about the centre-of-mass of B, where B is *not spinning*. These results should be compared and contrasted with Example Three of Lecture 17, which concerned a rocket-propelled road vehicle situated on Earth. Although there are no explicit spins involved in that example, the vehicle could be considered as orbiting the centre-of-mass of Earth, whist Earth is slowly spinning. In this sense, Example Three resembles a case in which the body B is spinning while the orbiting particle P is not spinning and so is essentially complementary to the cases discussed above.

In Example Three, it is the rotating Earth which gives the stationary vehicle its angular momentum in its constrained orbit about the Earth's centre. With this angular momentum, the vehicle's natural free orbit is much closer to Earth's centre and it is this difference between its constrained orbit and its natural orbit which gives the vehicle its weight. When the vehicle is set in motion either with or against Earth's rotation, its constrained angular momentum is changed, so that its weight is changed. Hence, the vehicle example also differs from the situation discussed in this lecture in the sense that the vehicle example investigates 'attraction' and 'repulsion' of an orbiting particle due to *changes* in angular momentum, whereas in this lecture, we have compared the 'attraction' and 'repulsion' of an orbiting particle when spinning and not spinning, for the *same* orbital angular momentum.

The effect of changes to the total kinetic energy on a natural, force-free circular orbit for fixed orbital angular momentum, may be compared and contrasted to the relationship between $\mathcal{K}$ and the 'gravitational factor' $\mathcal{G}$ with which $\mathcal{K}$ is associated. In the simplest case of an axially symmetric

particle P in a natural circular orbit about a body B, it follows by equations (7) and (8), and equation (17) of Lecture 18, *i.e.*,

$$\mathcal{G} = \frac{r_s v_s^2}{M} \qquad (10)$$

where M is the mass of the body B, that

$$\mathcal{G} = \frac{r_s v_s^2}{M} = \frac{L v_s}{mM} = \frac{L}{m^{3/2} M}(2\mathcal{K})^{1/2} \qquad (11)$$

so that for constant m, M and L, $\mathcal{G}$ is proportional to $\mathcal{K}^{1/2}$. Hence, $\mathcal{G}$ increases with increasing $\mathcal{K}$, in accordance with the fact that for constant L, as $\mathcal{K}$ increases, P is 'more attracted' to B, and as $\mathcal{K}$ decreases, P is 'less attracted' to B.

It is to be noted that we have only studied natural circular orbits of axially symmetric particles here. For natural elliptical orbits in general, there is no simple relationship between the orbital parameters and $\mathcal{K}$. In particular, r_s is not proportional to $1/\mathcal{K}^{1/2}$. Indeed, $\mathcal{K}$, r_s and v_s will not be constant. Nevertheless, the cases discussed here show that in POAMS, with fixed orbital angular momentum, the spin of a particle affects its relative 'attraction' towards a centre-of-mass in a way it cannot according to Newton's thesis.

We continue this lecture with the first of two numerical examples involving the effect of spin on orbital motion.

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3. Example One

Let us first consider the simplest example of the effect of spin on orbital motion, namely that of a satellite P in a natural circular orbit about Earth B. It should be appreciated from the outset that the figures and calculations quoted in this and the following examples can only be approximate, and that the primary intention is to give qualitative results together with the order of magnitude of predicted detectable effects. In this example, suppose that P follows a *polar orbit* around Earth, passing over each pole at a distance of 400 miles, *i.e.*, approximately 6.437376×10^5 metres,

above Earth's surface, the polar radius of Earth from its centre being approximately 6.356752×10^6 metres. The radius of the satellite's orbit at that height above Earth's surface, relative to Earth's centre, is then

$$r_0 \cong 7.000490 \times 10^6 \text{ m}.$$

Let the mass of this satellite be m kg. Suppose that, for the moment, P does not spin. Since P follows a natural (force-free) circular orbit, the orbital speed, v_0, of P is given in accordance with equation (8) of Lecture 16 $[v^2 = \frac{GM}{r}]$, so that

$$v_0 = \left(\frac{GM}{r_0}\right)^{1/2} \cong 7.562053 \times 10^3 \text{ m s}^{-1}$$

where $G \cong 6.6743015 \times 10^{-11}$ m^3 s^{-2} kg^{-1} and the Earth's mass, M, is calculated in POAMS to be $M \cong 5.997939 \times 10^{24}$ kg. Then the magnitude of P's orbital angular momentum, is from equation (7) of Lecture 16, $[L = mvr]$,

$$L = mv_0r_0 \cong 5.293808m \times 10^{10} \text{ kg m}^2\text{s}^{-1},$$

and its orbital kinetic energy is,

$$K_0 = \frac{1}{2}mv_0^2 \cong 2.859232m \times 10^7 \text{ kg m}^2\text{s}^{-2}$$

Suppose now that the satellite P, treated as a solid spherical ball, spins while it orbits Earth in its unconstrained natural circular orbit. Let us suppose that the diameter of the rotor is 3.81 cm and that it spins at 10,000 revolutions per minute. These figures are chosen here since these are respectively the diameter and spin rate of a quartz rotor used in the Gravity Probe B experiment, launched in 2004. The radius of P is then 1.905×10^{-2} metres. It follows by equation (8) of Lecture 18 $[K_S = \frac{1}{2}I\omega^2]$ and the formula for the moment of inertia of a solid sphere, *i.e.*, equation (4) of Lecture 18, $[I = \frac{2}{5}mr^2]$ that the spin kinetic energy, K_s, of any spinning sphere of uniform mass m is given by

$$K_s = \frac{1}{2}I\omega^2 = \frac{1}{5}m(r\omega)^2 = \frac{1}{5}mv^2 \qquad (12)$$

where I is the moment of inertia of the sphere, r is its radius, ω is its angular speed about its axis of rotation and v is its (equatorial) spin speed. Using equation (12), the spin kinetic energy for the spherical rotor in question is then

$$K_s \cong \frac{m}{5} \times \left(0.01905 \times 2\pi \times \frac{10000}{60}\right)^2 \cong 79.593425m \text{ kg m}^2\text{s}^{-2}$$

Suppose that the rotor spins in the same direction and in the same plane as its orbit. It then follows by equations (3) [$\mathcal{K} = K_o + K_s$] and (7) [$v_s^2 = \frac{2}{m}\mathcal{K}$] that the rotor's orbital speed, v_s, in its natural orbit accounting for effects due to its spin, is given by,

$$v_s^2 = \frac{2}{m}\mathcal{K} = \frac{2}{m}(K_o + K_s) \Rightarrow v_s \cong 7.562063 \times 10^3 \text{ m s}^{-1}$$

Then, by equation (8), the radius, r_s, of the rotor's natural circular orbit while spinning, for conserved L, is

$$r_s = \frac{L}{mv_s} \cong 7.000480 \times 10^6 \text{ m}$$

In this case, $r_s < r_0$, with $r_0 - r_s$ approximately 10 metres. Hence, POAMS predicts that the spinning rotor will follow a natural orbit approximately 10 metres *closer* to Earth than if it were not spinning. On the other hand, if the rotor spins counter to, but again in the same plane as its orbital motion, then it follows by equation (4) [$\mathcal{K} = |K_o - K_s|$], equation (7) and equation (8), that $r_s > r_0$, with $r_s - r_0$ approximately 10 metres. In this case, POAMS predicts that the rotor will naturally orbit 10 metres *further away* from Earth than if it were not spinning.

In this example, it is the *kinetic energies* that are added and subtracted, and not the angular momentum, which is assumed constant (conserved). It is in this way that this example with the spinning rotor differs from that of the rocket-propelled road vehicle, *i.e.*, Example Three of Lecture 17, in which the angular momentum of the vehicle was added to, or subtracted from, that of Earth. However, in order to send the satellite into orbit, angular

momentum has to be added, in which case the effect on the satellite is the same as that of the vehicle. On the other hand, once settled into its free orbit, the satellite and Earth form a paired and balanced constant angular momentum system, the effects of which are described by the preceding example.

Now suppose that the spin angular momentum vector of P is at right angles to P's orbital angular momentum vector. In this case, it follows by equations (6) [$K^2 = K_o^2 + K_s^2$] and (7) [$v_s^2 = \frac{2}{m}K$] that the orbital speed of the rotor is given by

$$v_s^2 = \frac{2}{m}K = \frac{2}{m}(K_o^2 + K_s^2)^{1/2} \Rightarrow v_s \cong 7.562052935280 \times 10^3 \text{ m s}^{-1}$$

In this scenario, the orbital radius of the rotor's natural orbit is then

$$r_s \cong 7.000489999986 \times 10^6 \text{ m}$$

In this case, $r_0 - r_s$ is approximately 0.000014 metres, *i.e.*, 0.014 millimetres, so that the effect of spin on the orbit will be negligible!

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4. Gravity Probe B

An example of a spinning satellite as described in Example One was Gravity Probe B. This was a satellite-based experiment containing four spinning gyroscopes, designed to test two unverified predictions of General Relativity: the *geodetic effect* and *frame dragging*. It was headed by a team from Stanford University and funded by NASA. The satellite was launched in April 2004 and the experiment lasted for 16 months. The gyroscopes contained spinning spheres, with the spin directions permanently pointing towards the guide star IM Pegasi. We understand that this meant that the spin directions were at right angles to the orbital angular momentum and so, as discussed in the preceding example, there would have been an almost negligible effect on the orbital radius caused by spin. Incidentally, each sphere had a mass m = 7.52916642 grams. In the experiment, the rotors were permanently spinning, so that it would not have been possible in any case to detect a difference between the natural orbit in the spin and no-spin

cases. Also, there was a precession effect due to the spin orientations, which would have affected any POAMS predictions.

In the next lecture, I will provide another numerical example involving spin, this time based on a terrestrial experiment involving weight gain or loss. We shall then move on to discuss the central role of spin in POAMS at the quantum level.

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(*) $\|\underset{\sim}{L}+\underset{\sim}{S}\|^2 = (\underset{\sim}{L}+\underset{\sim}{S})\cdot(\underset{\sim}{L}+\underset{\sim}{S})$

$= \underset{\sim}{L}\cdot\underset{\sim}{L} + 2\underset{\sim}{L}\cdot\underset{\sim}{S} + \underset{\sim}{S}\cdot\underset{\sim}{S}$

$\therefore \|\underset{\sim}{L}+\underset{\sim}{S}\|^2 = L^2 + 2LS\cos\theta + S^2$ (*)

(**) In (*), $0 \leq \theta^\circ \leq 180^\circ$. For $0 \leq \theta^\circ \leq 90^\circ$, $0 \leq \cos\theta \leq 1$, with $\underline{\cos 0^\circ = 1}$, $\underline{\cos 90^\circ = 0}$.

When $\theta^\circ = 0^\circ$, $\|\underset{\sim}{L}+\underset{\sim}{S}\|^2 = L^2 + 2LS + S^2$ by (*)

$= (L+S)^2$

Hence, $\|\underset{\sim}{L}+\underset{\sim}{S}\| = L+S$.

When $\theta^\circ = 90^\circ$, $\|\underset{\sim}{L}+\underset{\sim}{S}\| = L^2+S^2$, giving eqn. (6).

When $\theta^\circ = 180^\circ$, $\underline{\cos\theta^\circ = -1}$ and,

$\|\underset{\sim}{L}+\underset{\sim}{S}\|^2 = L^2 - 2LS + S^2$ by (*)

$= (L-S)^2$

Hence, $\|\underset{\sim}{L}+\underset{\sim}{S}\| = |\underset{\sim}{L}-\underset{\sim}{S}|$

For $90^\circ \leq \theta^\circ \leq 180^\circ$, $-1 \leq \cos\theta \leq 0$.
